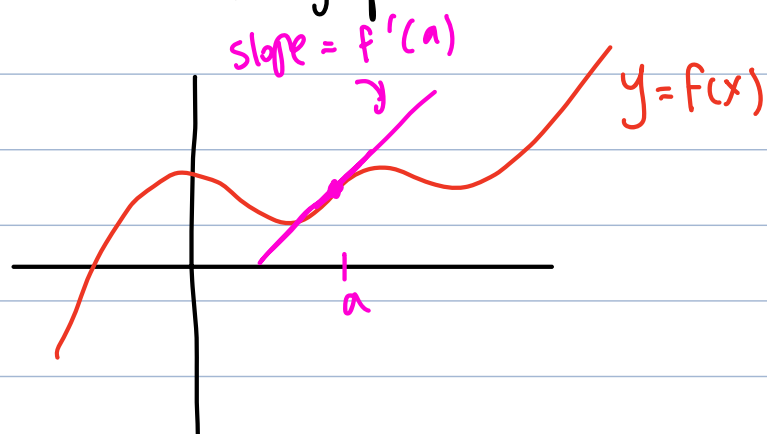
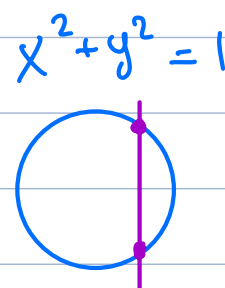


Calc 1 Lecture 11: Implicit differentiation

So far, we know how to find tangent lines to curves that are graphs of functions



What about more general curves? e.g. circles

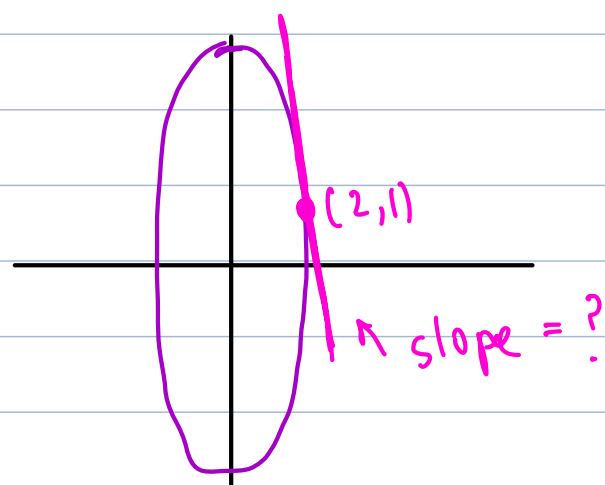


Example: What is the tangent line to the ellipse

$2x^2 + y^2 = 9$ at the point $(x, y) = (2, 1)$?

(can double check
 $(2, 1)$ is on the curve)

$$2 \cdot 2^2 + 1^2 = 9 \quad \checkmark$$



Approach #1: Try to write y in terms of x
(not implicit)

$$2x^2 + y^2 = 9$$

$$y^2 = 9 - 2x^2 \Rightarrow y = \pm \sqrt{9 - 2x^2}$$

Note: y is not a function of x .

But it is a function of x if we focus near our point.

Since $(2,1)$ is positive, focus on $y = \sqrt{9 - 2x^2}$

$$y' = \frac{d}{dx} (\sqrt{9 - 2x^2}) = \frac{1}{2\sqrt{9 - 2x^2}} \cdot \underbrace{\frac{d}{dx} (9 - 2x^2)}_{-4x}$$

$$= \frac{1}{2\sqrt{9 - 2x^2}} \cdot (-4x) = y \cdot \left\{ \frac{-2x}{\sqrt{9 - 2x^2}} \right\}$$

$$\text{At } (2,1): \text{ plug in } x=2, y=1: y' = \frac{-2 \cdot 2}{\sqrt{9 - 2 \cdot 2^2}} = \boxed{-4}$$

But this type of approach doesn't always work

Example: Find y' if $x^3 + y^3 = 6xy$

if you try isolating y ... you get stuck

Approach #2: implicit differentiation!

Idea: we can implicitly treat y as a function of x

$$y = f(x) \quad (\text{at least near the point of interest})$$

without explicitly writing down f .

The method of implicit differentiation:

To find y' (given an equation relating x and y):

- Step 0: Write down the equation relating x and y
- Step 1: Differentiate both sides with respect to x , while treating y implicitly as a function of x .
- Step 2: Solve for y' (in terms of both x and y)
- Step 3: If evaluating at a specific point, plug in the values of x and y .

Example Solve for y' if $2x^2 + y^2 = 9$

using implicit differentiation, and evaluate at $(x, y) = (2, 1)$

Solution:

step 0: Write down the equation relating x and y

$$2x^2 + y^2 = 9$$

step 1: Differentiate both sides with respect to x

$$\frac{d}{dx}(2x^2 + y^2) = \frac{d}{dx}(9)$$

$$\underbrace{\frac{d}{dx}(2x^2)}_{4x} + \frac{d}{dx}(y^2) = 0$$

Caution: $\frac{d}{dx}(y^2) \neq 2y$

$$\frac{d}{dx}(y^2) = \underbrace{\frac{d}{dy}(y^2)}_{2y} \cdot \underbrace{\frac{dy}{dx}}_{y'} = 2y \cdot y'$$

$$\boxed{4x + 2y \cdot y' = 0} \quad (*)$$

Step 2: Solve for y'

$$4x + 2y \cdot y' = 0 \Rightarrow 2y \cdot y' = -4x$$

$$\Rightarrow y' = \frac{-4x}{2y} = \boxed{-\frac{2x}{y}}$$

step 3: plug in specific values of x and y

$$\text{at } x=2, y=1: y' = \frac{-2 \cdot 2}{1} = \boxed{-4}$$

Note: we can skip step 2 if we only are interested in a specific point.

Quick exercises:

Treat y implicitly as a function of x .

Differentiate the following expressions with respect to x :

$$a) \ y^3 \quad \frac{d}{dx}(y^3) = \frac{d}{dy}(y^3) \cdot \frac{dy}{dx} = \underline{3y^2 \cdot y'}$$

$$b) \ xy \quad \frac{d}{dx}(xy) = \underbrace{\frac{d}{dx}(x)}_1 \cdot y + x \cdot \underbrace{\frac{d}{dx}(y)}_{y'} = \underline{y + x \cdot y'}$$

$$c) \ xy^2 \quad \frac{d}{dx}(xy^2) = \underbrace{\frac{d}{dx}(x)}_1 \cdot y^2 + x \cdot \underbrace{\frac{d}{dx}(y^2)}_{\frac{d}{dy}(y^2) \cdot y' = 2y \cdot y'} = \underline{y^2 + x \cdot 2y \cdot y'}$$

$$d) \ \frac{x}{y} \quad \frac{d}{dx}\left(\frac{x}{y}\right) = \frac{\underbrace{\frac{d}{dx}(x)}_1 \cdot y - x \cdot \underbrace{\frac{d}{dx}(y)}_{y'}}{y^2} = \underline{\frac{y - xy'}{y^2}}$$

$$e) \ \sin(x+y)$$

chain rule: $\frac{d}{dx}(\sin(\underline{x+y})) = \cos(x+y) \cdot \underbrace{\frac{d}{dx}(x+y)}_{\substack{\frac{d}{dx}(x) + \frac{d}{dx}(y) \\ " \\ 1 + y'}}$

$\hookrightarrow \boxed{\cos(x+y) \cdot (1+y')}$

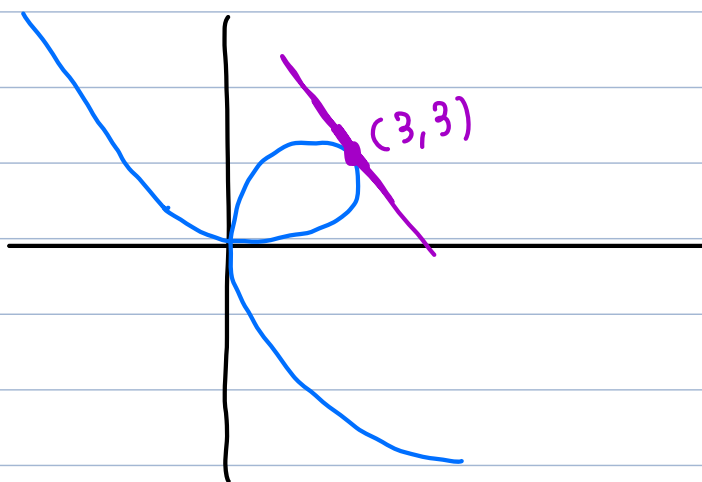
Example: Consider the "folium of Descartes",
leaf

$$x^3 + y^3 = 6xy$$

a) Find y' for points on this curve.

b) Find the tangent line at (3,3).

← double check this
is on curve



Descartes
"I think therefore
I am"

Solution:

a)

step 0: Write down the equation relating x and y

$$x^3 + y^3 = 6xy$$

step 1: Differentiate both sides (w.r.t. x)

$$\frac{d}{dx}(x^3 + y^3) = \frac{d}{dx}(6xy)$$

$$\underbrace{\frac{d}{dx}(x^3)}_{3x^2} + \underbrace{\frac{d}{dx}(y^3)}_{3y^2 \cdot y'} = 6 \underbrace{\frac{d}{dx}(xy)}_{y + xy'}$$

Step 2: solve for y'

$$3x^2 + 3y^2 \cdot y' = 6y + 6xy'$$

$$3y^2 \cdot y' - 6xy' = 6y - 3x^2$$

$$y'(3y^2 - 6x) = 6y - 3x^2 \Rightarrow$$

$$\boxed{y' = \frac{6y - 3x^2}{3y^2 - 6x}}$$

b)

Step 3: plug in specific values of x and y

$$(x, y) = \underline{(3, 3)}: \quad y' = \frac{6 \cdot 3 - 3 \cdot 3^2}{3 \cdot 3^2 - 6 \cdot 3} = \frac{-9}{9} = \underline{\underline{-1}}$$

tangent line:

$$y = y_0 + m(x - x_0)$$

$$m = -1, \quad (x_0, y_0) = (3, 3) \Rightarrow y = 3 - 1(x - 3)$$

$$\Rightarrow y = 3 - x + 3 \Rightarrow \boxed{y = 6 - x}$$

Note: Sometimes a curve may fail to have a tangent line at a given point (or have a vertical tangent line). In these cases y' is undefined.

(e.g. example above has undefined tangent at $(0, 0)$)

Followup question: at what points does the Folium

$(x^3 + y^3 = 6xy)$ have a horizontal tangent line? $y' = 0$

$$y' = \frac{6y - 3x^2}{3y^2 - 6x} = 0 \Rightarrow 6y - 3x^2 = 0 \Rightarrow 6y = 3x^2 \Rightarrow \boxed{y = \frac{x^2}{2}}$$

$\Rightarrow$ combine with equation of curve (substitute back in)

$$x^3 + \left(\frac{x^2}{2}\right)^3 = 6x \cdot \frac{x^2}{2}$$

$$\underline{x^3} + \frac{x^6}{8} = 3x^3 \Rightarrow \frac{x^6}{8} = 2x^3 \Rightarrow x^6 = 16x^3$$
$$x^3 = 16$$
$$x = \sqrt[3]{16}$$

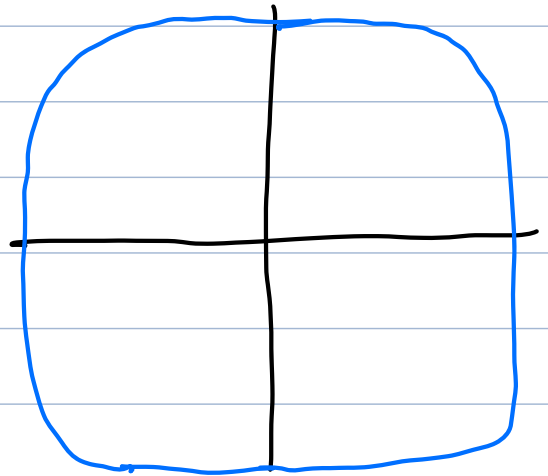
$$y = \frac{x^2}{2}$$

Example Use implicit differentiation to show that the tangent line to a circle at point P is perpendicular to the segment $\overline{OP}$ (O = origin)

Example: if $x^4 + y^4 = 1$

find y''

Idea: first use implicit diff.
to find y'



Then differentiate again (w.r.t. x)

$$x^4 + y^4 = 1$$

$$\frac{d}{dx}(x^4) + \frac{d}{dx}(y^4) = \frac{d}{dx}(1)$$

$$4x^3 + 4y^3 \cdot y' = 0$$

$$4y^3 \cdot y' = -4x^3 \Rightarrow$$

$$y' = \frac{-x^3}{y^3}$$

Take derivative again:

$$y'' = -\frac{d}{dx}\left(\frac{x^3}{y^3}\right) = -\frac{\overbrace{\frac{d}{dx}(x^3)}^{3x^2} \cdot y^3 - x^3 \cdot \overbrace{\frac{d}{dx}(y^3)}^{3y^2 \cdot y'}}{(y^3)^2}$$

$$= \frac{3x^2 y^3 - 3x^3 y^2 y'}{y^6}$$

$$= \frac{3x^2 y^3 - 3x^3 y^2 \left(\frac{-x^3}{y^3}\right)}{y^6}$$

Note: if we are only trying to solve for y' (or y'' or y''' etc.) at a specific point

we can be "lazy" and not isolate y' until plugging in the point.

Example: Consider the curve

$$xy + x^2 - y^3 = 1$$

Find y'' at the point $(x,y) = (1,0)$.

$$xy + x^2 - y^3 = 1 \quad \left) \quad \frac{d}{dx}$$

$$\underbrace{\frac{d}{dx}(xy)}_{\frac{d}{dx}(x) \cdot y + x \cdot \frac{dy}{dx}} + \underbrace{\frac{d}{dx}(x^2)}_{2x} - \underbrace{\frac{d}{dx}(y^3)}_{3y^2 \cdot y'} = \frac{d}{dx}(1)$$
$$y + xy' + 2x - 3y^2 y' = 0$$

① $y + xy' + 2x - 3y^2 y' = 0$

Plug in $(x,y) = (1,0)$:

$$\cancel{0} + 1 \cdot y' + 2 - 3 \cdot \cancel{0^2} \cdot y' = 0 \Rightarrow \boxed{y' = -2}$$

② apply $\frac{d}{dx}$ to equation ①

$$\frac{d}{dx}(y) + \frac{d}{dx}(xy') + \frac{d}{dx}(2x) - 3 \frac{d}{dx}(y^2 y') = 0$$

$$y' + \left(1 \cdot y' + x \cdot \frac{d}{dx}(y') \right) + 2 - 3 \left(\frac{d}{dx}(y^2) y' + y^2 \cdot \frac{d}{dx}(y') \right) = 0$$

y'' $2y \cdot y'$ y''

$$\textcircled{2} \quad \frac{y' + y'}{2y'} + x y'' + 2 - 3(2y \cdot (y')^2 + y^2 \cdot y'') = 0$$

Evaluate at our point: $x=1, y=0, y'=-2$

$$2 \cdot (-2) + 1 \cdot y'' + 2 - 3(2 \cdot 0 \cdot (-2)^2 + 0^2 \cdot y'') = 0$$

$$-4 + y'' + 2 = 0 \Rightarrow \boxed{y'' = 2}$$

Using implicit to differentiate inverse functions

Example: $f(x) = \sin^{-1}(x)$

$$f'(x) = ?$$

$$y = \sin^{-1}x \iff x = \sin y$$

$$x = \sin y \quad \left. \begin{array}{l} \text{apply } \frac{d}{dx} \end{array} \right\}$$

$$\underbrace{\frac{d}{dx}(x)}_1 = \underbrace{\frac{d}{dx}(\sin y)}_{\cos(y) \cdot y'} \Rightarrow 1 = \cos(y) \cdot y' \Rightarrow \boxed{y' = \frac{1}{\cos(y)}}$$

Question: what is y' as a function of x ?

$$y = \sin^{-1}x \Rightarrow y' = \frac{1}{\cos(\sin^{-1}x)}$$

Know: $\sin y = x$

$$\cos y = ?$$

Approach 1: use trig identity $\cos^2 y + \underbrace{\sin^2 y}_{x^2} = 1$

$$\cos^2 y = 1 - x^2$$

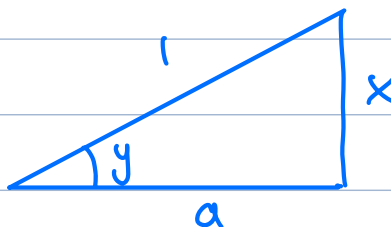
$$\cos y = \sqrt{1-x^2}$$

$$y' = \frac{1}{\cos(y)} = \boxed{\frac{1}{\sqrt{1-x^2}}}$$

Approach 2: draw a triangle

$$\sin y = x$$

$$\frac{\text{opp}}{\text{hyp.}} = \frac{x}{1}$$



$$\text{Pythagoras: } a^2 + x^2 = 1$$

$$\Rightarrow a = \sqrt{1-x^2}$$

$$\cos y = \frac{\text{adj}}{\text{hyp}} = \frac{a}{1} = \boxed{\sqrt{1-x^2}}$$

$$\leftarrow y' = \frac{1}{\cos y}$$

In conclusion:

$$\boxed{\frac{d}{dx} (\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}}$$

domain of $\sin^{-1} x$: $[-1, 1]$

domain of $\frac{d}{dx} (\sin^{-1} x)$: $1-x^2 > 0 \Rightarrow x^2 < 1$
 $-1 < x < 1$

$(-1, 1)$

Derivatives of inverse trig functions

$$\frac{d}{dx} (\sin^{-1}(x)) = \frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx} (\cos^{-1}(x)) = -\frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx} (\tan^{-1}(x)) = \frac{1}{1+x^2}$$

$$\frac{d}{dx} (\csc^{-1}(x)) = -\frac{1}{x\sqrt{x^2-1}}$$

$$\frac{d}{dx} (\sec^{-1}(x)) = \frac{1}{x\sqrt{x^2-1}}$$

$$\frac{d}{dx} (\cot^{-1}(x)) = -\frac{1}{1+x^2}$$

Derivative of $\ln(x)$

$$y = \ln x \iff \underbrace{x = e^y}_{\frac{d}{dx}}$$

$$\underbrace{\frac{d}{dx}(x)}_1 = \underbrace{\frac{d}{dx}(e^y)}_{e^y \cdot y'}$$

$$1 = e^y \cdot y' \Rightarrow y' = \frac{1}{\underbrace{e^y}_x} = \frac{1}{x}$$

To summarize:

$$\boxed{\frac{d}{dx}(\ln x) = \frac{1}{x}}$$

Example: find $\frac{d}{dx}(\sin^{-1}(\underbrace{\ln x}_u))$

$$u = \ln x$$

$$\frac{du}{dx} = \frac{1}{x}$$

$$\begin{aligned} \frac{d}{dx}(\sin^{-1} u) &= \frac{d}{du}(\sin^{-1} u) \cdot \frac{du}{dx} \\ &\stackrel{\text{chain rule}}{=} \frac{1}{\sqrt{1-u^2}} \cdot \frac{1}{x} \end{aligned}$$

$$= \boxed{\frac{1}{\sqrt{1-(\ln x)^2}} \cdot \frac{1}{x}}$$

Back to example: $f(x) = x^x$

convert to base e .

$$x = e^{\ln x}$$

raise to the x

$$\underline{x^x} = (e^{\ln x})^x = \underline{e^{\ln x \cdot x}}$$

$$\frac{d}{dx}(x^x) = \frac{d}{dx}(e^{\ln x \cdot x}) = e^{\ln x \cdot x} \cdot \underbrace{\frac{d}{dx}(\ln x \cdot x)}_{\substack{\frac{d}{dx}(\ln x) \cdot x + \ln x \cdot \frac{d}{dx}(x) \\ \frac{1}{x} \cdot x + \ln(x)}}$$

$$= \underbrace{e^{\ln x \cdot x}}_{x^x} \cdot (1 + \ln(x))$$

$$\boxed{\frac{d}{dx}(x^x) = x^x \cdot (1 + \ln(x))}$$